Case Study 07: Supermarket Retail Billing with Discounts and Taxes

1. Problem Statement

Develop a Python program to compute an itemized point-of-sale customer receipt for a 3-item shopping cart. The program reads the unit price and purchase quantity for three items, validates that prices and quantities are non-negative, halts immediately upon detecting invalid input, computes the gross subtotal, applies an automated 10% promotional discount if the subtotal reaches or exceeds Rs. 1000, levies a 5% Goods and Services Tax (GST) on the discounted taxable amount, and prints an itemized billing summary.

2. Objectives

- To read and validate numerical user inputs across multiple items using a controlled iteration loop.
- To implement a for...else loop control structure coupled with an early break statement to catch input errors cleanly.
- To implement conditional discount logic using ternary conditional expressions.
- To compute tax liabilities on net taxable amounts and output formatted financial metrics rounded to two decimal places.

3. Assumptions

- The billing transaction processes exactly 3 distinct merchandise items.
- Unit price is entered as a floating-point number, and quantity is entered as an integer.
- Any negative price or quantity triggers an immediate error message and terminates the transaction without generating a final bill.
- A 10% discount applies if and only if the cumulative subtotal is greater than or equal to Rs. 1000.
- A flat 5% GST is computed strictly on the post-discount taxable amount (taxable = subtotal - discount).

4. Algorithm

- **Step 1:** Start the program.

- **Step 2:** Initialize subtotal = 0.

- **Step 3:** For each item index i from 1 to 3:

    - Prompt and read price as a float and quantity as an integer.

    - If price < 0 or quantity < 0: display "Invalid item values", break out of the loop, and proceed to Step 9.

    - Add the item total to the accumulator: subtotal += price * quantity.

- **Step 4:** If the loop finishes all 3 items without breaking (else block of loop):

    - Compute discount = subtotal * 0.10 if subtotal >= 1000, else set discount = 0.

    - Compute net taxable base: taxable = subtotal - discount.

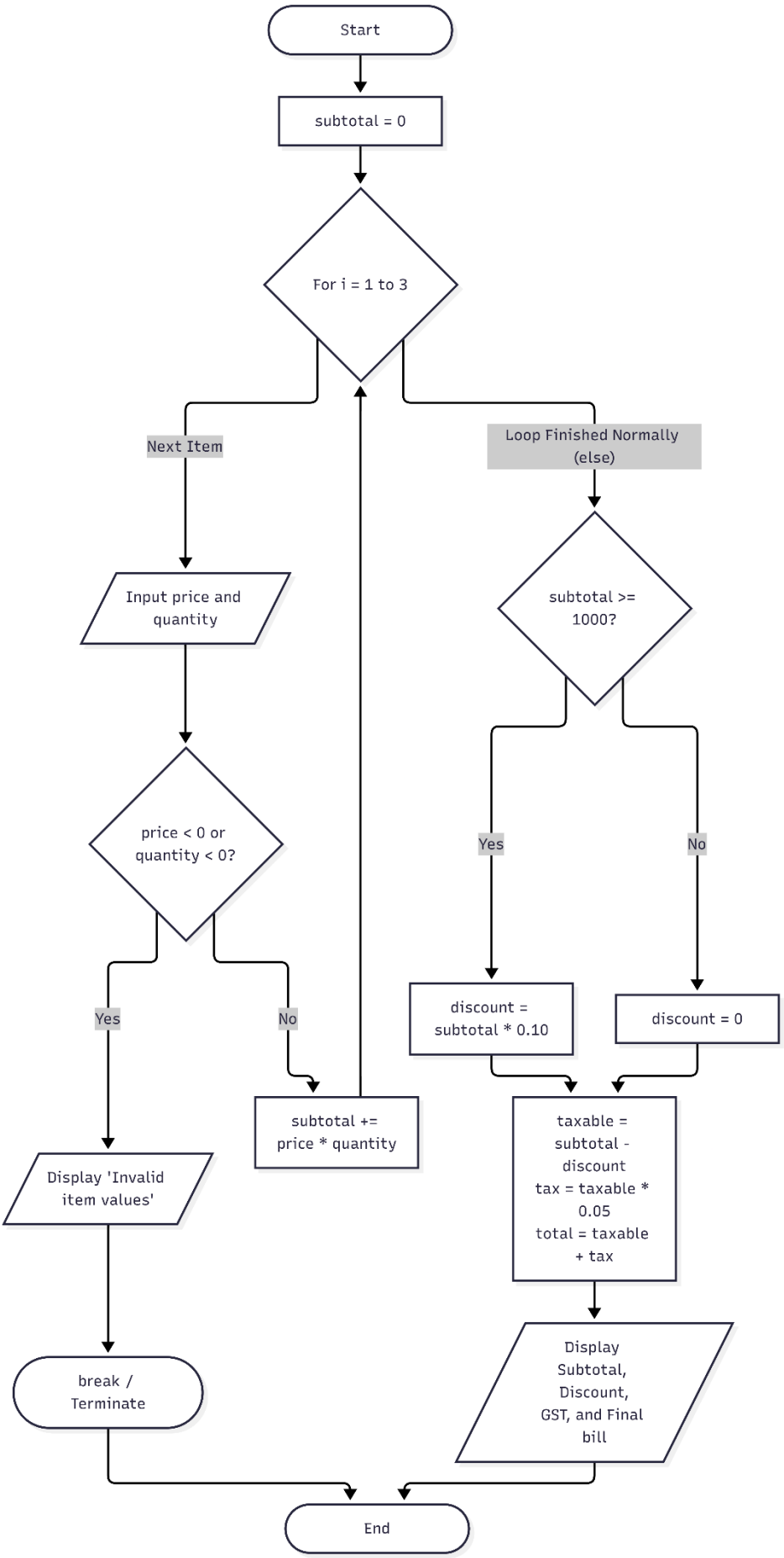
    - Compute sales tax: tax = taxable * 0.05.

    - Compute total bill: total = taxable + tax.

- **Step 5:** Print the formatted subtotal, discount, GST, and final bill.

- **Step 6:** Stop the program.

5. Flowchart Logic / Diagram


Start
subtotal = 0
For i = 1 to 3
Next Item
Input price and quantity
price < 0 or quantity < 0?
Yes
No
Display 'Invalid item values'
break / Terminate
subtotal += price * quantity
Loop Finished Normally (else)
subtotal >= 1000?
Yes
No
discount = subtotal * 0.10
discount = 0
taxable = subtotal - discount
tax = taxable * 0.05
total = taxable + tax
Display Subtotal, Discount, GST, and Final bill
End

6. Pseudocode

```
Plaintext
BEGIN
  SET subtotal = 0.0

  FOR i FROM 1 TO 3 DO
    PRINT "Enter price of item " + i + ": "
    READ price
    PRINT "Enter quantity: "
    READ quantity

    IF price < 0 OR quantity < 0 THEN
      PRINT "Invalid item values"
      BREAK
    END IF

    subtotal = subtotal + (price * quantity)
  ELSE
    IF subtotal >= 1000 THEN
      discount = subtotal * 0.10
    ELSE
      discount = 0.0
    END IF
```

```
    taxable = subtotal - discount
    tax = taxable * 0.05
    total = taxable + tax

    PRINT "Subtotal: Rs.", ROUND(subtotal, 2)
    PRINT "Discount: Rs.", ROUND(discount, 2)
    PRINT "GST: Rs.", ROUND(tax, 2)
    PRINT "Final bill: Rs.", ROUND(total, 2)
  END FOR
END
```

7. Python Source Code

Python

```
# Supermarket Retail Billing with Discounts and Taxes

# Initialize billing accumulator
subtotal = 0

# Capture pricing and quantities for 3 items
for i in range(1, 4):
  price = float(input("Enter price of item " + str(i) + ": "))
  quantity = int(input("Enter quantity: "))

  # Validate inputs
```

```
    if price < 0 or quantity < 0:
        print("Invalid item values")
        break
    subtotal += price * quantity
else:
    # Evaluate 10% discount threshold (Rs. 1000)
    discount = subtotal * 0.10 if subtotal >= 1000 else 0
    taxable = subtotal - discount
    tax = taxable * 0.05
    total = taxable + tax

    # Display receipt summary
    print("Subtotal: Rs.", round(subtotal, 2))
    print("Discount: Rs.", round(discount, 2))
    print("GST: Rs.", round(tax, 2))
    print("Final bill: Rs.", round(total, 2))
```

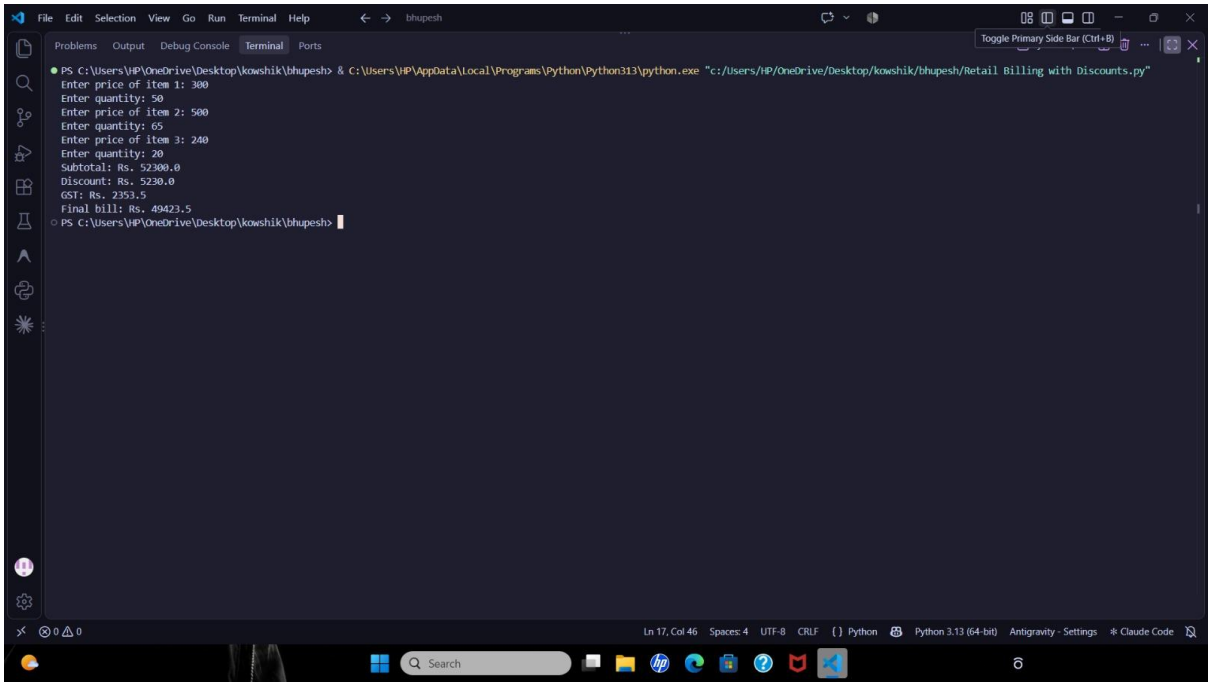
8. Test Cases

Test Case ID	**Input Sequence (Price, Qty)**	**Expected Output**	**Actual Output**	**Status**
TC_01 (Discount Applied)	Item 1: 300, 2	Subtotal: Rs. 1100.0	Subtotal: Rs. 1100.0	Pass

Test Case ID	Input Sequence (Price, Qty)	Expected Output	Actual Output	Status
	Item 2: 200, 1 Item 3: 150, 2	Discount: Rs. 110.0 GST: Rs. 49.5 Final bill: Rs. 1039.5	Discount: Rs. 110.0 GST: Rs. 49.5 Final bill: Rs. 1039.5	
TC_02 (No Discount Applied)	Item 1: 100, 2 Item 2: 50, 4 Item 3: 100, 1	Subtotal: Rs. 500.0 Discount: Rs. 0 GST: Rs. 25.0 Final bill: Rs. 525.0	Subtotal: Rs. 500.0 Discount: Rs. 0 GST: Rs. 25.0 Final bill: Rs. 525.0	Pass

Test Case ID	Input Sequence (Price, Qty)	Expected Output	Actual Output	Status
TC_03 (Negative Price)	Item 1: 250, 2 Item 2: -50, 1	Invalid item values	Invalid item values	Pass
TC_04 (Negative Quantity)	Item 1: 150, -2	Invalid item values	Invalid item values	Pass
TC_05 (Exact Discount Boundary)	Item 1: 500, 1 Item 2: 250, 2 Item 3: 0, 1	Subtotal: Rs. 1000.0 Discount: Rs. 100.0 GST: Rs. 45.0 Final bill: Rs. 945.0	Subtotal: Rs. 1000.0 Discount: Rs. 100.0 GST: Rs. 45.0 Final bill: Rs. 945.0	Pass

9. Output Screenshots



10. Conclusion

The Supermarket Retail Billing program was executed and verified successfully. The script demonstrates the application of Python's unique for...else loop construct, loop interruption with break, inline conditional validation, and chained financial arithmetic for discounts and taxation.